

Initiating Search

November 4, 2025, 11:42 AM

• Search:

CAS Lexicon Search:

Liebeskind-Srogl coupling reaction - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (123)	 References	View Results
Exported: Retrieved Related Reaction Results (553)	 Reactions	View Results

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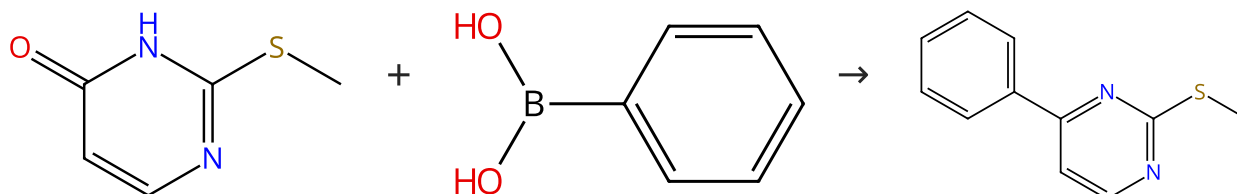
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Reactions (50)

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Scheme 1 (1 Reaction)

Steps: 1 Yield: 99%


 Suppliers (104)

 Suppliers (127)

 Suppliers (42)

31-179-CAS-10790891

Steps: 1 Yield: 99%

A general regioselective synthesis of 2,4-diarylpyrimidines from 2-thiouracil through two orthogonal cross-coupling reactions

By: Cernova, Miroslava; et al

Synlett (2012), 23(9), 1305-1308.

1.1 Reagents: Triethylamine, Phosphorus(1+), bromotri-1-pyrroli danyl-, (7-4)-, hexafluorophosphate(1-) (1:1)

Solvents: 1,4-Dioxane; 2 h, rt

1.2 Reagents: Potassium carbonate

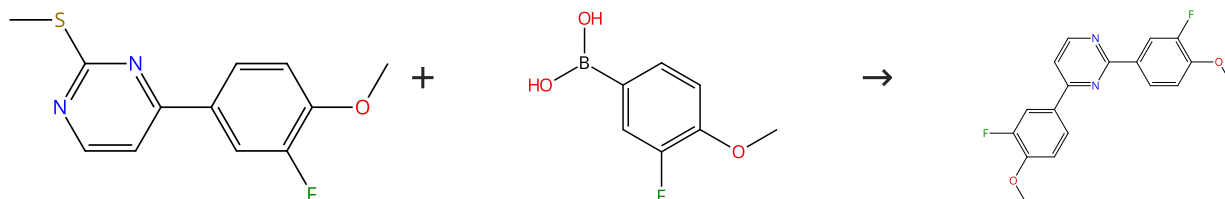
Catalysts: Dichlorobis(triphenylphosphine)palladium

Solvents: Water; 4 h, 100 °C

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 99%


 Suppliers (2)

 Suppliers (96)

 Supplier (1)

31-179-CAS-10451025

Steps: 1 Yield: 99%

A general regioselective synthesis of 2,4-diarylpyrimidines from 2-thiouracil through two orthogonal cross-coupling reactions

By: Cernova, Miroslava; et al

Synlett (2012), 23(9), 1305-1308.

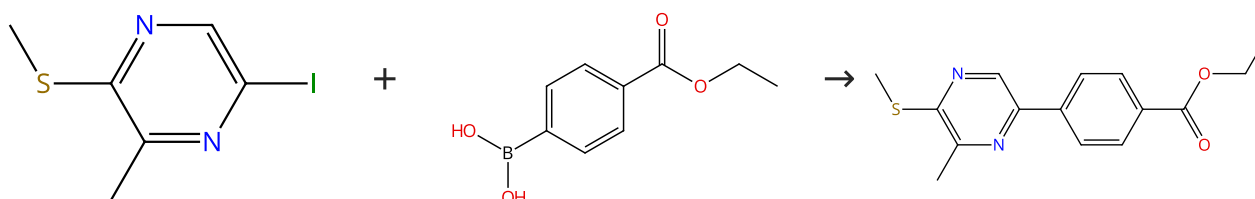
1.1 Catalysts: Tetrakis(triphenylphosphine)palladium, (2-Thiophe necarboxylato-κO²,κS¹)copper

Solvents: Tetrahydrofuran; 1 h, 100 °C

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 99%


 Suppliers (2)

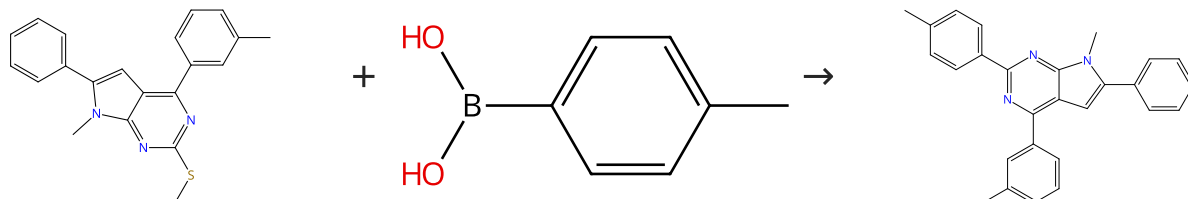
 Suppliers (94)

 Supplier (1)

31-179-CAS-5404297	Steps: 1 Yield: 99%	A Novel and Versatile Entry to Asymmetrically Substituted Pyrazines
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 10 min, 120 °C		By: Mehta, Vaibhav Pravinchandra; et al
Experimental Protocols		Journal of Organic Chemistry (2008), 73(6), 2382-2388.

Scheme 4 (1 Reaction)

Steps: 1 Yield: 99%

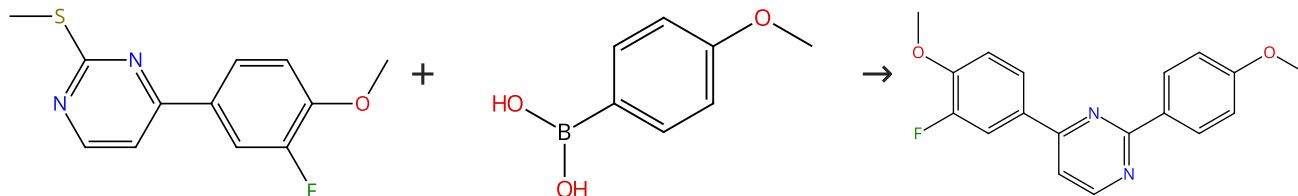


Suppliers (114)

31-179-CAS-21177728	Steps: 1 Yield: 99%	A Strategy for the Triarylation of Pyrrolopyrimidines by Using Microwave-Promoted Cross-Coupling Reactions
1.1 Catalysts: Tetrakis(triphenylphosphine)palladium, [2-(Hydroxy-κO)-3-methylbenzoato-κO]copper Solvents: Tetrahydrofuran; 1 h, 100 °C		By: Prieur, Vanessa; et al
Experimental Protocols		European Journal of Organic Chemistry (2015), 2015(29), 6547-6556.

Scheme 5 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (2)

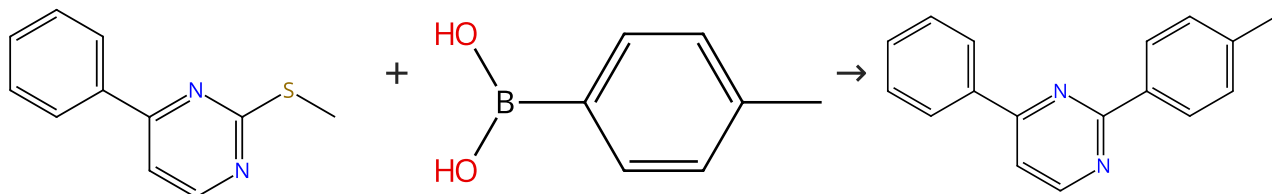
Suppliers (113)

Supplier (1)

31-179-CAS-8324631	Steps: 1 Yield: 99%	A general regioselective synthesis of 2,4-diarylpyrimidines from 2-thiouracil through two orthogonal cross-coupling reactions
1.1 Catalysts: Tetrakis(triphenylphosphine)palladium, (2-Thiophenecarboxylato-κO ² ,κS ¹)copper Solvents: Tetrahydrofuran; 1 h, 100 °C		By: Cernova, Miroslava; et al
Experimental Protocols		Synlett (2012), 23(9), 1305-1308.

Scheme 6 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (42)

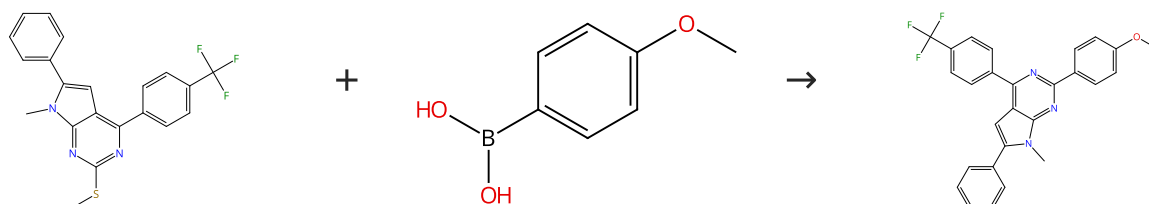
Suppliers (114)

Suppliers (8)

31-179-CAS-5920016	Steps: 1 Yield: 99%	A general regioselective synthesis of 2,4-diarylpyrimidines from 2-thiouracil through two orthogonal cross-coupling reactions By: Cernova, Miroslava; et al Synlett (2012), 23(9), 1305-1308.
1.1 Catalysts: Tetrakis(triphenylphosphine)palladium, (2-Thiophenecarboxylato- $\kappa O^2, \kappa S^1$)copper Solvents: Tetrahydrofuran; 1 h, 100 °C		
Experimental Protocols		

Scheme 7 (1 Reaction)

Steps: 1 Yield: 99%

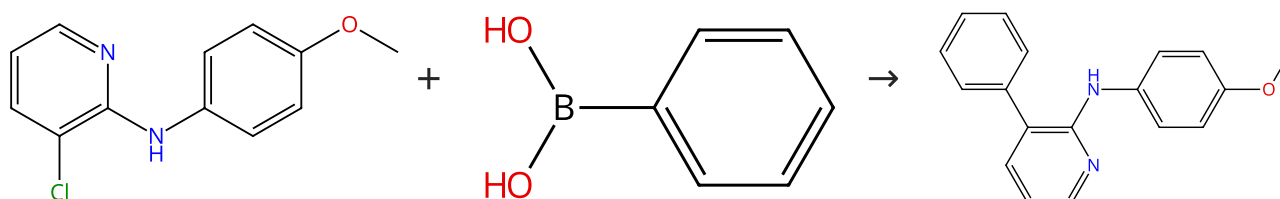


Suppliers (113)

31-179-CAS-21181777	Steps: 1 Yield: 99%	A Strategy for the Triarylation of Pyrrolopyrimidines by Using Microwave-Promoted Cross-Coupling Reactions By: Prieur, Vanessa; et al European Journal of Organic Chemistry (2015), 2015(29), 6547-6556.
1.1 Catalysts: Tetrakis(triphenylphosphine)palladium, [2-(Hydroxy- κO)-3-methylbenzoato- κO]copper Solvents: Tetrahydrofuran; 1 h, 100 °C		
Experimental Protocols		

Scheme 8 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (8)

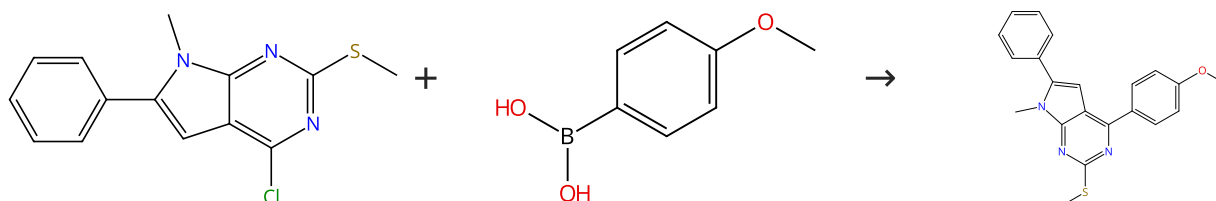
Suppliers (127)

Supplier (1)

31-179-CAS-14154826	Steps: 1 Yield: 99%	Regioselective Syntheses of 2,3-Substituted Pyridines by Orthogonal Cross-Coupling Strategies By: Koley, Moumita; et al European Journal of Organic Chemistry (2011), (10), 1972-1979, S1972/1-S1972/98.
1.1 Reagents: Potassium carbonate Catalysts: Palladium diacetate, X-Phos Solvents: Toluene; 14 h, 150 °C		
Experimental Protocols		

Scheme 9 (1 Reaction)

Steps: 1 Yield: 99%

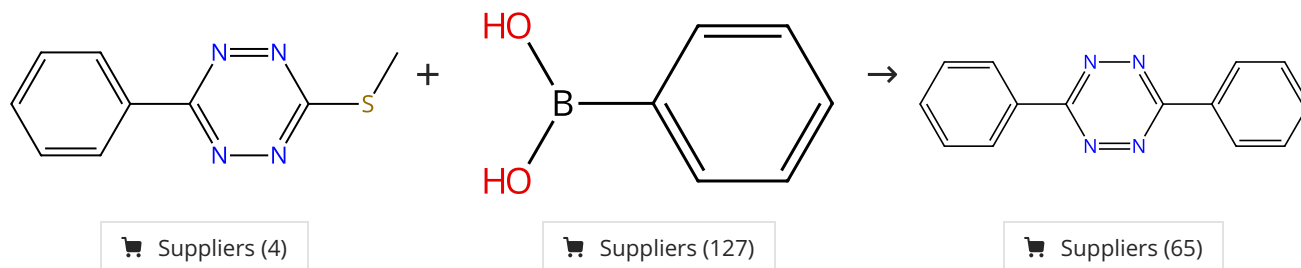


Suppliers (113)

31-179-CAS-14182937	Steps: 1 Yield: 99%	A Strategy for the Triarylation of Pyrrolopyrimidines by Using Microwave-Promoted Cross-Coupling Reactions
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,2-Dimethoxyethane, Water; 90 min, 100 °C		By: Prieur, Vanessa; et al
Experimental Protocols		European Journal of Organic Chemistry (2015), 2015(29), 6547-6556.

Scheme 10 (1 Reaction)

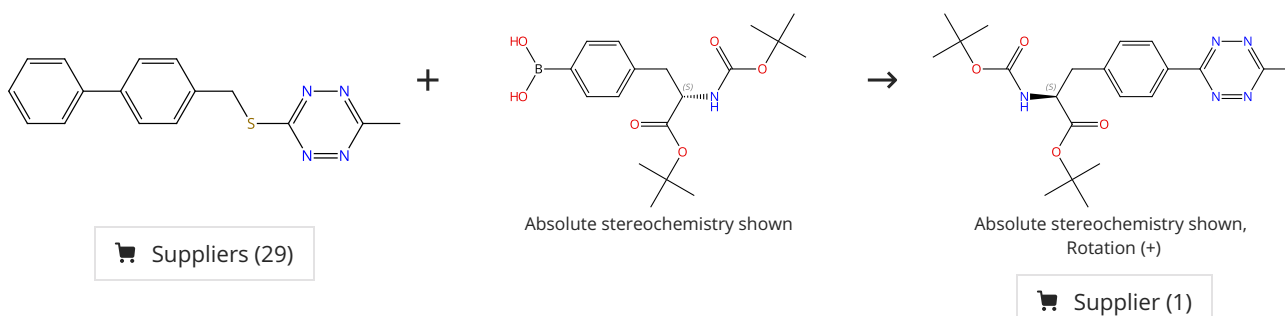
Steps: 1 Yield: 99%



31-179-CAS-21158600	Steps: 1 Yield: 99%	Installation of Minimal Tetrazines through Silver-Mediated Liebeskind-Srogl Coupling with Arylboronic Acids
1.1 Reagents: Silver oxide (Ag ₂ O) Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Dimethylformamide; 19 - 21 h, 60 °C		By: Lambert, William D.; et al
Experimental Protocols		Journal of the American Chemical Society (2019), 141(43), 17068-17074.

Scheme 11 (1 Reaction)

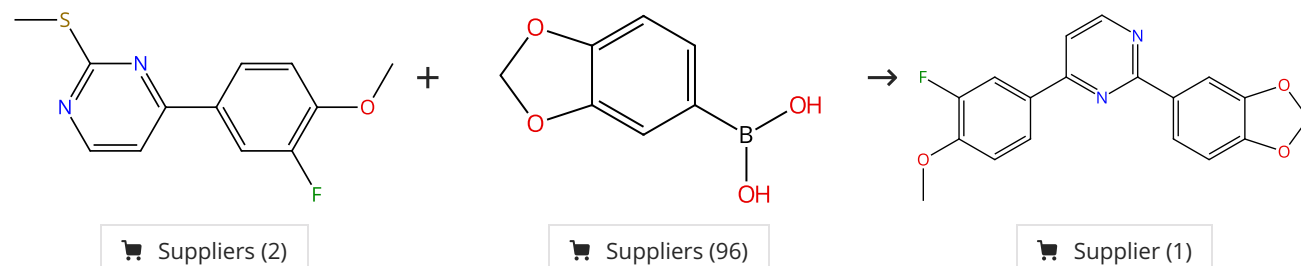
Steps: 1 Yield: 98%



31-179-CAS-22103176	Steps: 1 Yield: 98%	Installation of Minimal Tetrazines through Silver-Mediated Liebeskind-Srogl Coupling with Arylboronic Acids
1.1 Reagents: Silver oxide (Ag ₂ O) Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Dimethylformamide; 20 h, 60 °C		By: Lambert, William D.; et al
Experimental Protocols		Journal of the American Chemical Society (2019), 141(43), 17068-17074.

Scheme 12 (1 Reaction)

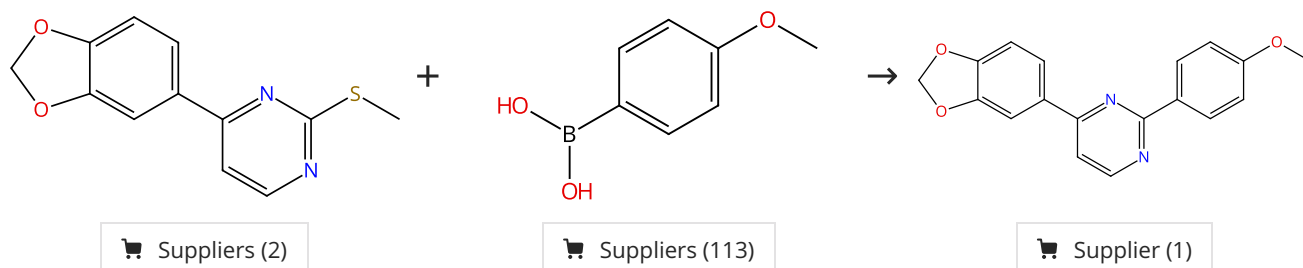
Steps: 1 Yield: 98%



31-179-CAS-12558098 Steps: 1 Yield: 98% 1.1 Catalysts: Tetrakis(triphenylphosphine)palladium, (2-Thiophene-5-carboxylato- $\kappa O^2, \kappa S^1$)copper Solvents: Tetrahydrofuran; 1 h, 100 °C Experimental Protocols	A general regioselective synthesis of 2,4-diarylpyrimidines from 2-thiouracil through two orthogonal cross-coupling reactions By: Cernova, Miroslava; et al Synlett (2012), 23(9), 1305-1308.
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Scheme 13 (1 Reaction)

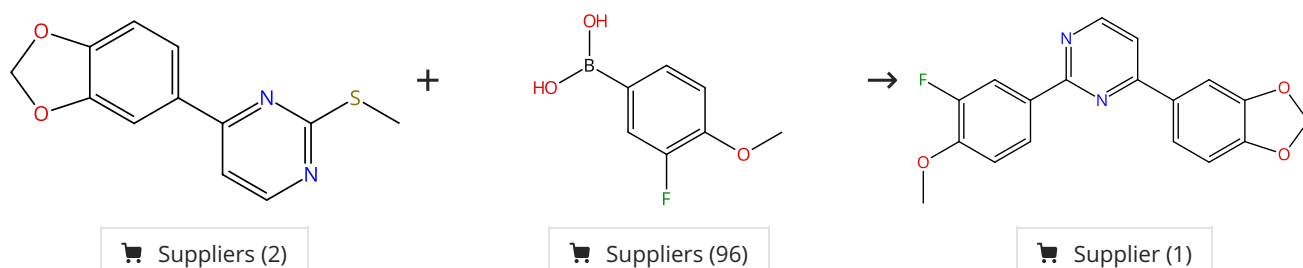
Steps: 1 Yield: 98%



31-179-CAS-1317965 Steps: 1 Yield: 98% 1.1 Catalysts: Tetrakis(triphenylphosphine)palladium, (2-Thiophene-5-carboxylato- $\kappa O^2, \kappa S^1$)copper Solvents: Tetrahydrofuran; 1 h, 100 °C Experimental Protocols	A general regioselective synthesis of 2,4-diarylpyrimidines from 2-thiouracil through two orthogonal cross-coupling reactions By: Cernova, Miroslava; et al Synlett (2012), 23(9), 1305-1308.
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Scheme 14 (1 Reaction)

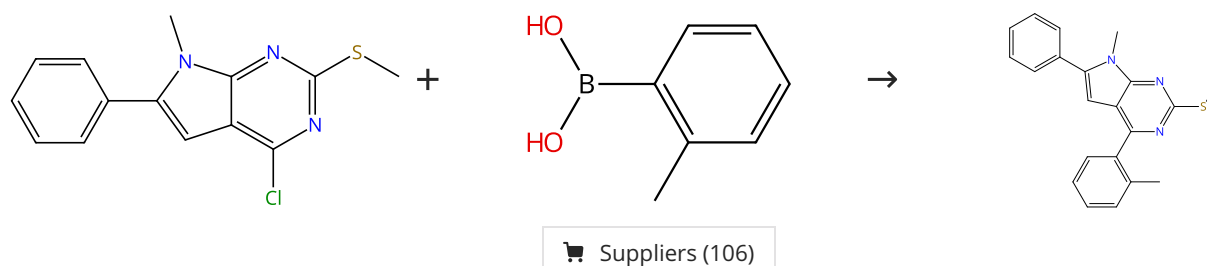
Steps: 1 Yield: 98%



31-179-CAS-3457664 Steps: 1 Yield: 98% 1.1 Catalysts: Tetrakis(triphenylphosphine)palladium, (2-Thiophene-5-carboxylato- $\kappa O^2, \kappa S^1$)copper Solvents: Tetrahydrofuran; 1 h, 100 °C Experimental Protocols	A general regioselective synthesis of 2,4-diarylpyrimidines from 2-thiouracil through two orthogonal cross-coupling reactions By: Cernova, Miroslava; et al Synlett (2012), 23(9), 1305-1308.
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Scheme 15 (1 Reaction)

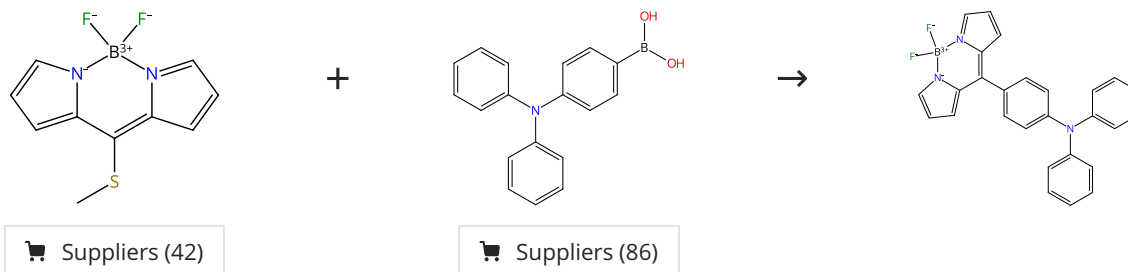
Steps: 1 Yield: 98%



31-179-CAS-2186945	Steps: 1 Yield: 98%	A Strategy for the Triarylation of Pyrrolopyrimidines by Using Microwave-Promoted Cross-Coupling Reactions
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,2-Dimethoxyethane, Water; 90 min, 100 °C		By: Prieur, Vanessa; et al
Experimental Protocols		European Journal of Organic Chemistry (2015), 2015(29), 6547-6556.

Scheme 16 (1 Reaction)

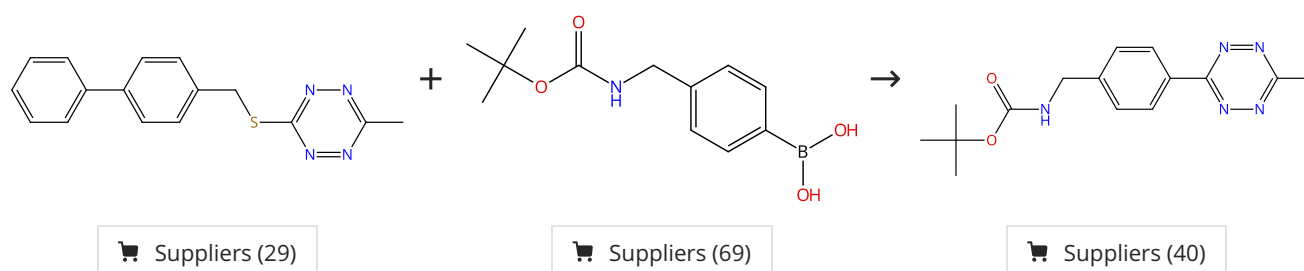
Steps: 1 Yield: 98%



31-179-CAS-18109300	Steps: 1 Yield: 98%	Novel meso-Polyarylamine-BODIPY Hybrids: Synthesis and Study of Their Optical Properties
1.1 Reagents: (2-Thiophenecarboxylato- $\kappa O^2, \kappa S^1$)copper Catalysts: Tris(2-furyl)phosphine, Tris(dibenzylideneacetone) dipalladium Solvents: Tetrahydrofuran; 55 °C		By: Lager, Erik; et al
Experimental Protocols		Journal of Organic Chemistry (2009), 74(5), 2053-2058.

Scheme 17 (1 Reaction)

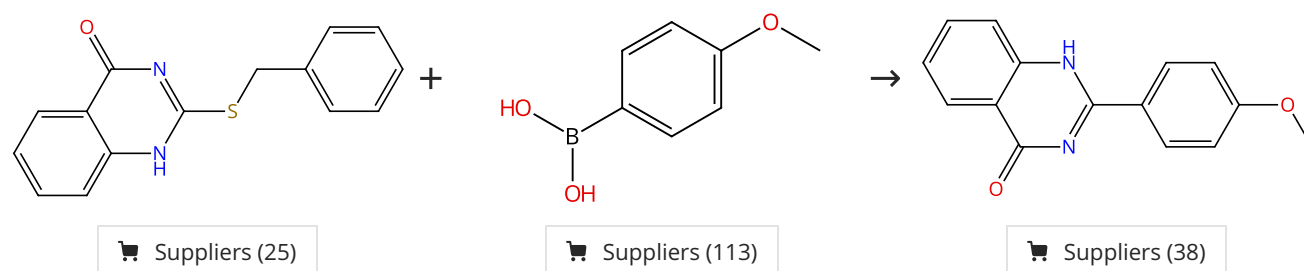
Steps: 1 Yield: 97%



31-179-CAS-22121953	Steps: 1 Yield: 97%	Installation of Minimal Tetrazines through Silver-Mediated Liebeskind-Srogl Coupling with Arylboronic Acids
1.1 Reagents: Silver oxide (Ag2O) Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Dimethylformamide; 19 - 21 h, 60 °C		By: Lambert, William D.; et al
Experimental Protocols		Journal of the American Chemical Society (2019), 141(43), 17068-17074.

Scheme 18 (1 Reaction)

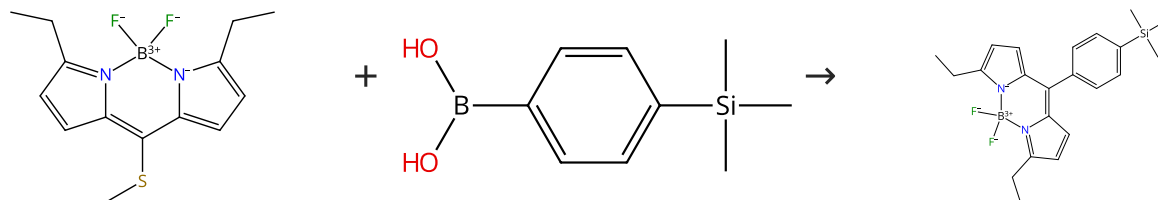
Steps: 1 Yield: 97%



31-179-CAS-21255151 Steps: 1 Yield: 97% 1.1 Catalysts: [2-(Hydroxy-κO)-3-methylbenzoato-κO]copper Solvents: Tetrahydrofuran; 10 min, rt 1.2 Catalysts: Tetrakis(triphenylphosphine)palladium; 1 h, 100 °C Experimental Protocols	Conversion of 2-thioxo-2,3-dihydroquinazolin-4(1H)-ones to N (3)-unsubstituted 2-(HET)arylquinazolin-4(3H)-ones by copper-mediated pd-catalysed cross-coupling reactions By: Krisciuniene, Vilija; et al Heterocycles (2016), 93(1), 150-163.
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Scheme 19 (1 Reaction)

Steps: 1 Yield: 97%

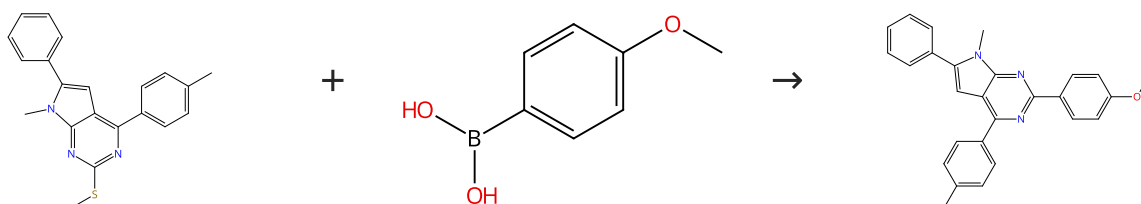


Suppliers (80)

31-179-CAS-18144386 Steps: 1 Yield: 97% 1.1 Reagents: (2-Thiophenecarboxylato-$\kappa O^2, \kappa S^1$)copper Catalysts: Tris(2-furyl)phosphine, Tris(dibenzylideneacetone) dipalladium Solvents: Tetrahydrofuran; 5 min; 0.75 h, 55 °C Experimental Protocols	Scope and Limitations of the Liebeskind-Srogl Cross-Coupling Reactions Involving the Biellmann BODIPY By: Betancourt-Mendiola, Lourdes; et al Journal of Organic Chemistry (2015), 80(11), 5771-5782.
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Scheme 20 (1 Reaction)

Steps: 1 Yield: 97%

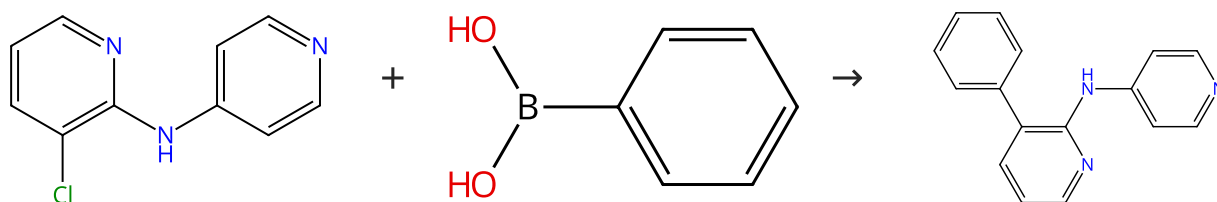


Suppliers (113)

31-179-CAS-21179748 Steps: 1 Yield: 97% 1.1 Catalysts: Tetrakis(triphenylphosphine)palladium, [2-(Hydroxy-κO)-3-methylbenzoato-κO]copper Solvents: Tetrahydrofuran; 1 h, 100 °C Experimental Protocols	A Strategy for the Triarylation of Pyrrolopyrimidines by Using Microwave-Promoted Cross-Coupling Reactions By: Prieur, Vanessa; et al European Journal of Organic Chemistry (2015), 2015(29), 6547-6556.
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Scheme 21 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (4)

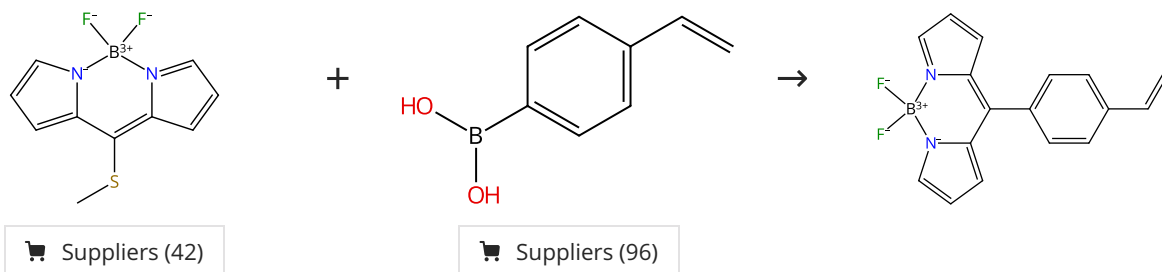
Suppliers (127)

Supplier (1)

31-179-CAS-8590113	Steps: 1 Yield: 97%	Regioselective Syntheses of 2,3-Substituted Pyridines by Orthogonal Cross-Coupling Strategies By: Koley, Moumita; et al European Journal of Organic Chemistry (2011), (10), 1972-1979, S1972/1-S1972/98.
1.1 Reagents: Potassium carbonate Catalysts: Palladium diacetate, X-Phos Solvents: Toluene; 14 h, 150 °C		
Experimental Protocols		

Scheme 22 (1 Reaction)

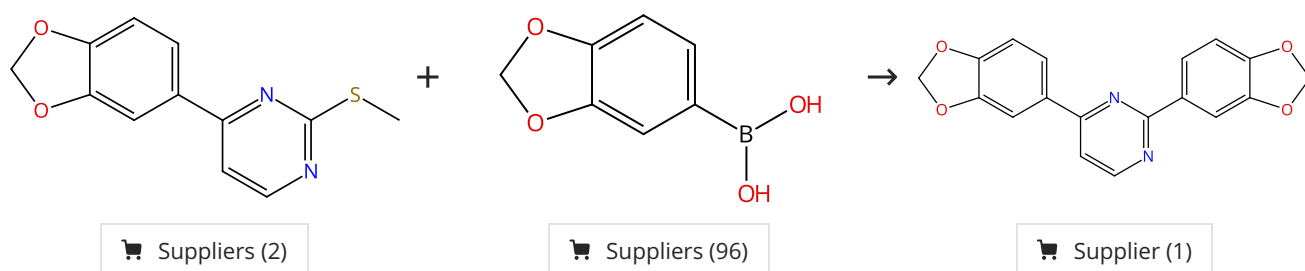
Steps: 1 Yield: 96%



31-179-CAS-18217526	Steps: 1 Yield: 96%	Simple, general, and efficient synthesis of meso-substituted borondipyrromethenes from a single platform By: Pena-Cabrera, Eduardo; et al Organic Letters (2007), 9(20), 3985-3988.
1.1 Catalysts: Tris(2-furyl)phosphine, Tris(dibenzylideneacetone) dipalladium, (2-Thiophenecarboxylato- $\kappa O^2, \kappa S^1$)copper Solvents: Tetrahydrofuran; 10 min, rt; 25 min, 55 °C		
Experimental Protocols		

Scheme 23 (1 Reaction)

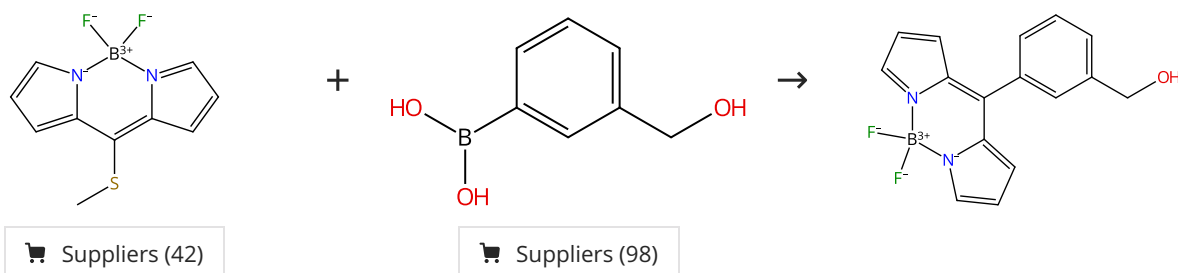
Steps: 1 Yield: 96%



31-179-CAS-5574800	Steps: 1 Yield: 96%	A general regioselective synthesis of 2,4-diarylpyrimidines from 2-thiouracil through two orthogonal cross-coupling reactions By: Cernova, Miroslava; et al Synlett (2012), 23(9), 1305-1308.
1.1 Catalysts: Tetrakis(triphenylphosphine)palladium, (2-Thiophenecarboxylato- $\kappa O^2, \kappa S^1$)copper Solvents: Tetrahydrofuran; 1 h, 100 °C		
Experimental Protocols		

Scheme 24 (2 Reactions)

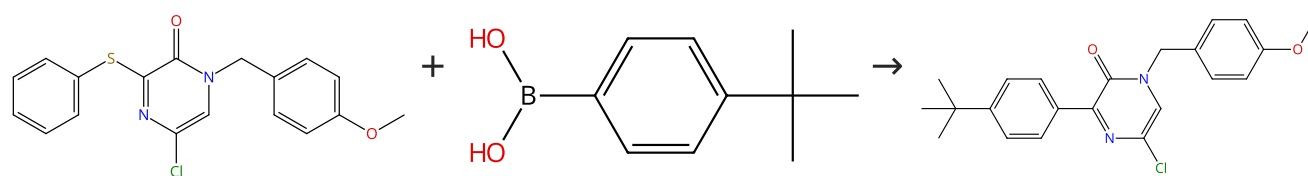
Steps: 1 Yield: 95-96%



31-179-CAS-18179729 Steps: 1 Yield: 96% 1.1 Reagents: (2-Thiophenecarboxylato-$\kappa O^2, \kappa S^1$)copper Catalysts: Tris(2-furyl)phosphine, Tris(dibenzylideneacetone) dipalladium Solvents: Tetrahydrofuran; 5 min; 1 h, 55 °C Experimental Protocols	Scope and Limitations of the Liebeskind-Srogl Cross-Coupling Reactions Involving the Biellmann BODIPY By: Betancourt-Mendiola, Lourdes; et al Journal of Organic Chemistry (2015), 80(11), 5771-5782.
31-179-CAS-18052796 Steps: 1 Yield: 95% 1.1 Reagents: (2-Thiophenecarboxylato-$\kappa O^2, \kappa S^1$)copper Catalysts: Tris(2-furyl)phosphine, Tris(dibenzylideneacetone) dipalladium Solvents: Tetrahydrofuran; 5 min; 5 min, 100 °C Experimental Protocols	Scope and Limitations of the Liebeskind-Srogl Cross-Coupling Reactions Involving the Biellmann BODIPY By: Betancourt-Mendiola, Lourdes; et al Journal of Organic Chemistry (2015), 80(11), 5771-5782.

Scheme 25 (1 Reaction)

Steps: 1 Yield: 96%

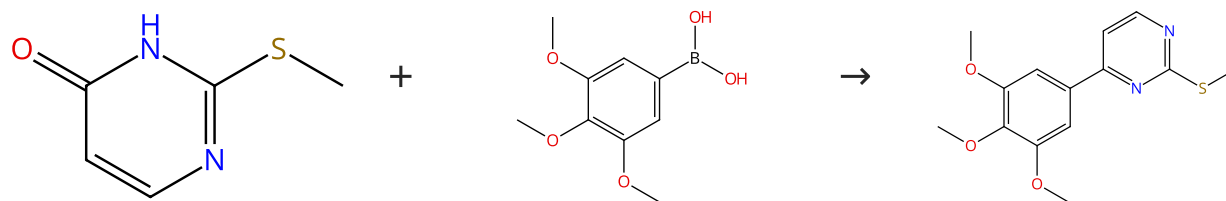


Suppliers (92)

31-179-CAS-1000742 Steps: 1 Yield: 96% 1.1 Reagents: (2-Thiophenecarboxylato-$\kappa O^2, \kappa S^1$)copper Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Tetrahydrofuran; 1 h, 35 °C Experimental Protocols	Palladium-catalyzed copper(I)-mediated cross-coupling of arylboronic acids and 2(1H)-pyrazinones facilitated by microwave irradiation with simultaneous cooling By: Singh, Brajendra Kumar; et al Organic & Biomolecular Chemistry (2007), 5(18), 2962-2965.
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Scheme 26 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (104)

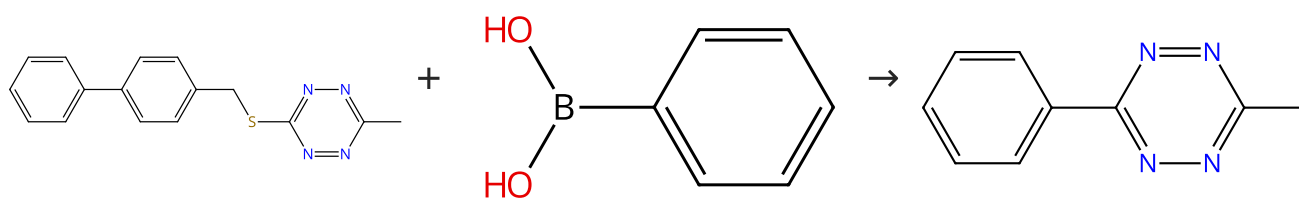
Suppliers (87)

Supplier (1)

31-179-CAS-3761969 Steps: 1 Yield: 96% 1.1 Reagents: Triethylamine, Phosphorus(1+), bromotri-1-pyrroli danyl-, (7-4)-, hexafluorophosphate(1-) (1:1) Solvents: 1,4-Dioxane; 2 h, rt 1.2 Reagents: Potassium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: Water; 4 h, 100 °C Experimental Protocols	A general regioselective synthesis of 2,4-diarylpyrimidines from 2-thiouracil through two orthogonal cross-coupling reactions By: Cernova, Miroslava; et al Synlett (2012), 23(9), 1305-1308.
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Scheme 27 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (29)

Suppliers (127)

Suppliers (19)

31-179-CAS-22142151

Steps: 1 Yield: 96%

Installation of Minimal Tetrazines through Silver-Mediated Liebeskind-Srogl Coupling with Arylboronic Acids

By: Lambert, William D.; et al

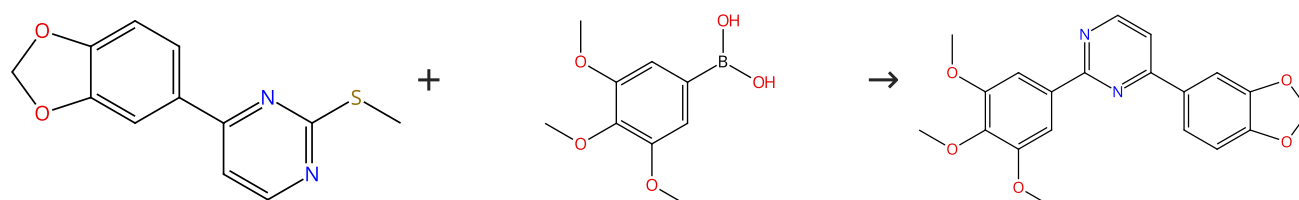
Journal of the American Chemical Society (2019), 141(43), 17068-17074.

1.1 **Reagents:** Silver oxide (Ag_2O)
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Dimethylformamide; 19 - 21 h, 60 °C

Experimental Protocols

Scheme 28 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (2)

Suppliers (87)

31-179-CAS-7721693

Steps: 1 Yield: 96%

A general regioselective synthesis of 2,4-diarylpyrimidines from 2-thiouracil through two orthogonal cross-coupling reactions

By: Cernova, Miroslava; et al

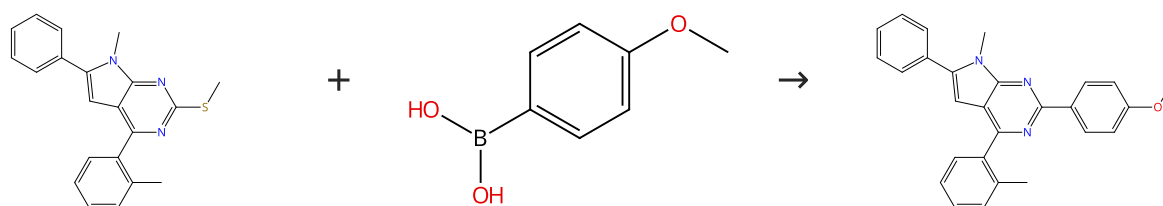
Synlett (2012), 23(9), 1305-1308.

1.1 **Catalysts:** Tetrakis(triphenylphosphine)palladium, (2-Thiophene-5-carboxylato- $\kappa\text{O}^2, \kappa\text{S}^1$)copper
Solvents: Tetrahydrofuran; 1 h, 100 °C

Experimental Protocols

Scheme 29 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (113)

31-179-CAS-21181813

Steps: 1 Yield: 95%

A Strategy for the Triarylation of Pyrrolopyrimidines by Using Microwave-Promoted Cross-Coupling Reactions

By: Prieur, Vanessa; et al

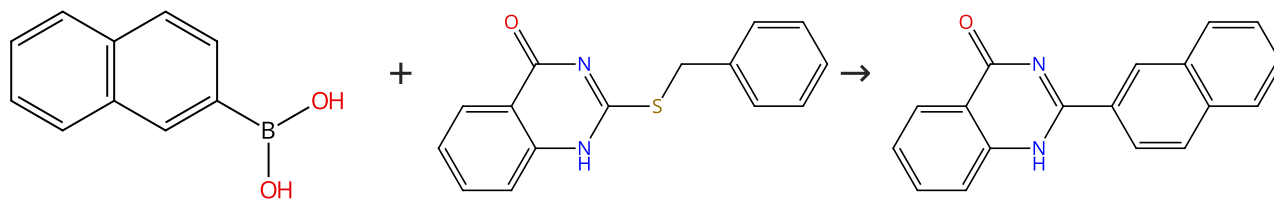
European Journal of Organic Chemistry (2015), 2015(29), 6547-6556.

1.1 **Catalysts:** Tetrakis(triphenylphosphine)palladium, [2-(Hydroxy- κO)-3-methylbenzoato- κO]copper
Solvents: Tetrahydrofuran; 1 h, 100 °C

Experimental Protocols

Scheme 30 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (102)

Suppliers (25)

Suppliers (10)

31-179-CAS-21201110

Steps: 1 Yield: 95%

Conversion of 2-thioxo-2,3-dihydroquinazolin-4(1H)-ones to N (3)-unsubstituted 2-(HET)arylquinazolin-4(3H)-ones by copper-mediated pd-catalysed cross-coupling reactions

1.1 Catalysts: [2-(Hydroxy-κO)-3-methylbenzoato-κO]copper
Solvents: Tetrahydrofuran; 10 min, rt

1.2 Catalysts: Tetrakis(triphenylphosphine)palladium; 48 h, reflux

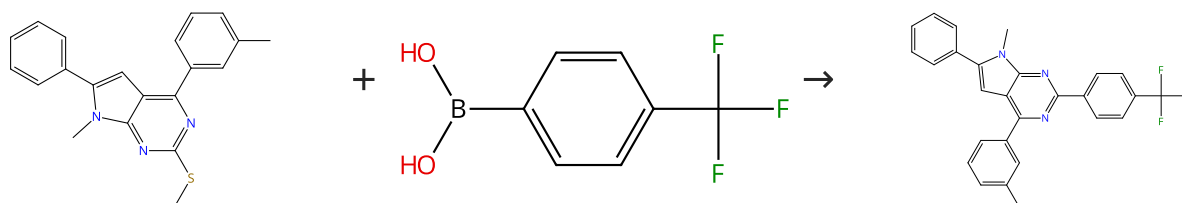
By: Krisciuniene, Vilija; et al

Heterocycles (2016), 93(1), 150-163.

Experimental Protocols

Scheme 31 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (111)

31-179-CAS-21179083

Steps: 1 Yield: 95%

A Strategy for the Triarylation of Pyrrolopyrimidines by Using Microwave-Promoted Cross-Coupling Reactions

1.1 Catalysts: Tetrakis(triphenylphosphine)palladium, [2-(Hydroxy-κO)-3-methylbenzoato-κO]copper
Solvents: Tetrahydrofuran; 1 h, 100 °C

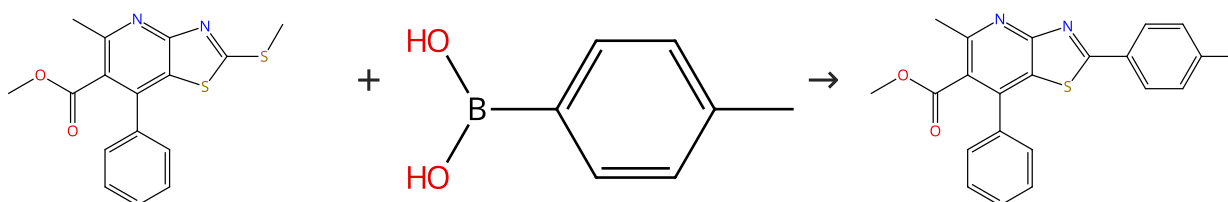
By: Prieur, Vanessa; et al

European Journal of Organic Chemistry (2015), 2015(29), 6547-6556.

Experimental Protocols

Scheme 32 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (114)

31-179-CAS-21055078

Steps: 1 Yield: 94%

New synthetic approach for the preparation of 2-aryl-thiazolo [4,5-b]pyridines via Liebeskind-Srogl reaction

1.1 Reagents: (2-Thiophenecarboxylato-κO²,κS¹)copper
Catalysts: Palladium
Solvents: Tetrahydrofuran; 12 h, 100 °C

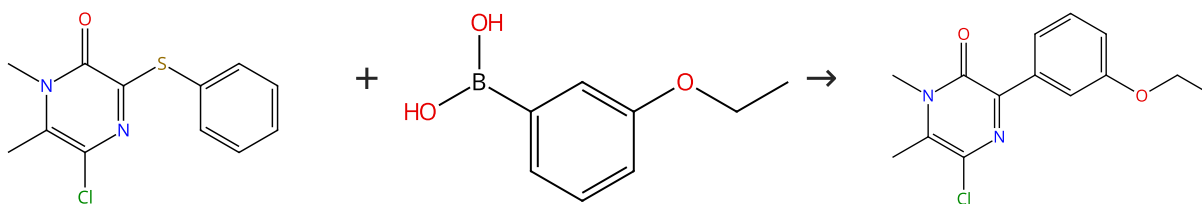
By: Peng, Yang-Qiu; et al

Chinese Chemical Letters (2015), 26(8), 1016-1018.

Experimental Protocols

Scheme 33 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (81)

31-179-CAS-14052488

Steps: 1 Yield: 94%

Palladium-catalyzed copper(I)-mediated cross-coupling of arylboronic acids and 2(1H)-pyrazinones facilitated by microwave irradiation with simultaneous cooling

By: Singh, Brajendra Kumar; et al

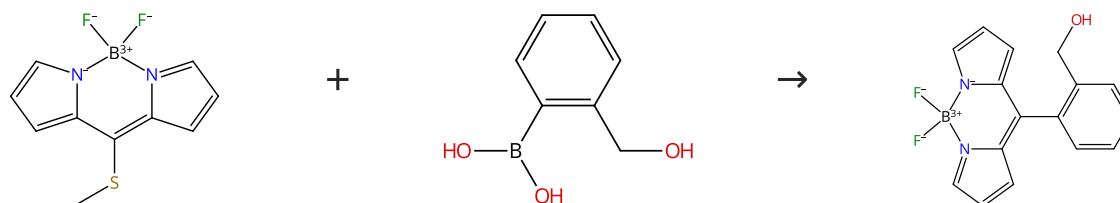
Organic & Biomolecular Chemistry (2007), 5(18), 2962-2965.

1.1 **Reagents:** (2-Thiophenecarboxylato- $\kappa O^2, \kappa S^1$)copper
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran; 1 h, 35 °C

Experimental Protocols

Scheme 34 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (42)

Suppliers (89)

31-179-CAS-18122127

Steps: 1 Yield: 94%

Scope and Limitations of the Liebeskind-Srogl Cross-Coupling Reactions Involving the Biellmann BODIPY

By: Betancourt-Mendiola, Lourdes; et al

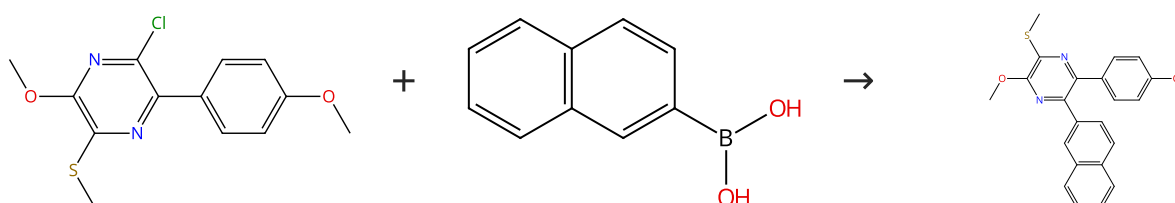
Journal of Organic Chemistry (2015), 80(11), 5771-5782.

1.1 **Reagents:** (2-Thiophenecarboxylato- $\kappa O^2, \kappa S^1$)copper
Catalysts: Tris(2-furyl)phosphine, Tris(dibenzylideneacetone) dipalladium
Solvents: Tetrahydrofuran; 5 min; 1 h, 55 °C

Experimental Protocols

Scheme 35 (1 Reaction)

Steps: 1 Yield: 94%



Supplier (1)

Suppliers (102)

Supplier (1)

31-179-CAS-12422892

Steps: 1 Yield: 94%

A Novel and Versatile Entry to Asymmetrically Substituted Pyrazines

By: Mehta, Vaibhav Pravinchandra; et al

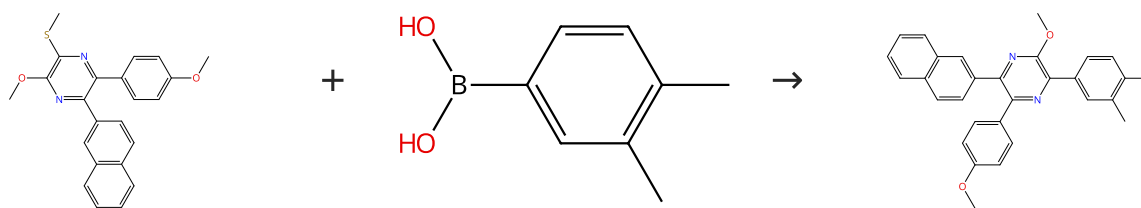
Journal of Organic Chemistry (2008), 73(6), 2382-2388.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 10 min, 120 °C

Experimental Protocols

Scheme 36 (1 Reaction)

Steps: 1 Yield: 94%



Supplier (1)

Suppliers (93)

Supplier (1)

31-179-CAS-21057106

Steps: 1 Yield: 94%

A Novel and Versatile Entry to Asymmetrically Substituted Pyrazines

By: Mehta, Vaibhav Pravinchandra; et al

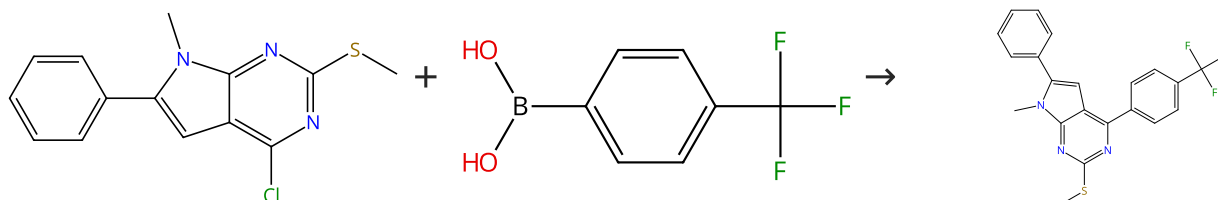
Journal of Organic Chemistry (2008), 73(6), 2382-2388.

1.1 **Catalysts:** Tetrakis(triphenylphosphine)palladium, (2-Thiophe necarboxylato- $\kappa O^2, \kappa S^1$)copper
Solvents: Tetrahydrofuran; 50 min, 110 °C

Experimental Protocols

Scheme 37 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (111)

31-179-CAS-7198826

Steps: 1 Yield: 94%

A Strategy for the Triarylation of Pyrrolopyrimidines by Using Microwave-Promoted Cross-Coupling Reactions

By: Prieur, Vanessa; et al

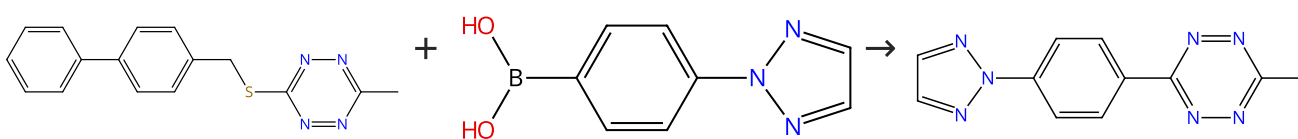
European Journal of Organic Chemistry (2015), 2015(29), 6547-6556.

1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,2-Dimethoxyethane, Water; 90 min, 100 °C

Experimental Protocols

Scheme 38 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (29)

Suppliers (5)

31-179-CAS-22104232

Steps: 1 Yield: 94%

Installation of Minimal Tetrazines through Silver-Mediated Liebeskind-Srogl Coupling with Arylboronic Acids

By: Lambert, William D.; et al

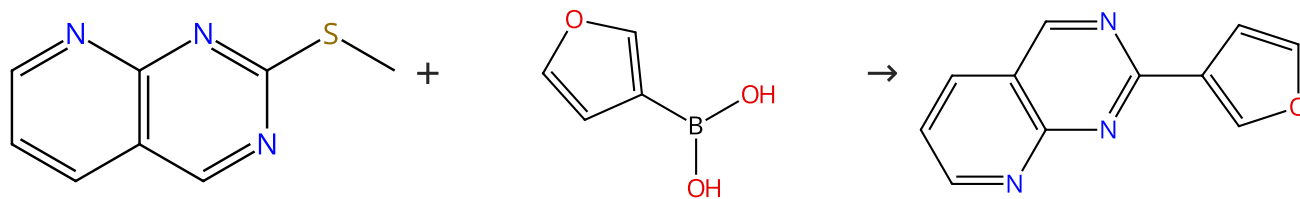
Journal of the American Chemical Society (2019), 141(43), 17068-17074.

1.1 **Reagents:** Silver oxide (Ag₂O)
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Dimethylformamide; 19 - 21 h, 60 °C

Experimental Protocols

Scheme 39 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (3)

Suppliers (101)

Suppliers (5)

31-179-CAS-22112137

Steps: 1 Yield: 93%

Photochemical route for the synthesis of novel 2- monosubstituted pyrido[2,3-d]pyrimidines by palladium-catalyzed cross-coupling reactions

By: Riadi, Yassine; et al

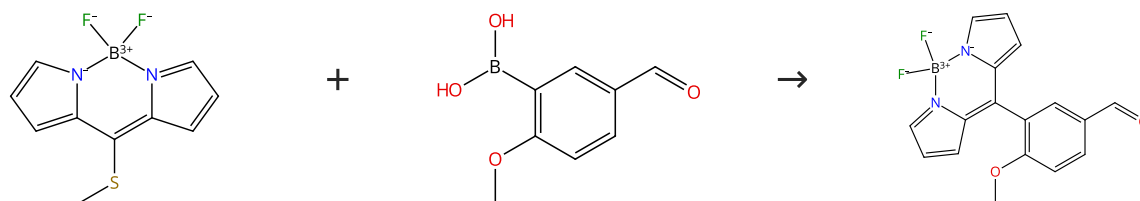
Chemical Papers (2018), 72(3), 697-701.

1.1 **Reagents:** (2-Thiophenecarboxylato- $\kappa O^2, \kappa S^1$)copper
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran; 3 h

Experimental Protocols

Scheme 40 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (42)

Suppliers (85)

31-179-CAS-18140226

Steps: 1 Yield: 93%

Scope and Limitations of the Liebeskind-Srogl Cross-Coupling Reactions Involving the Biellmann BODIPY

By: Betancourt-Mendiola, Lourdes; et al

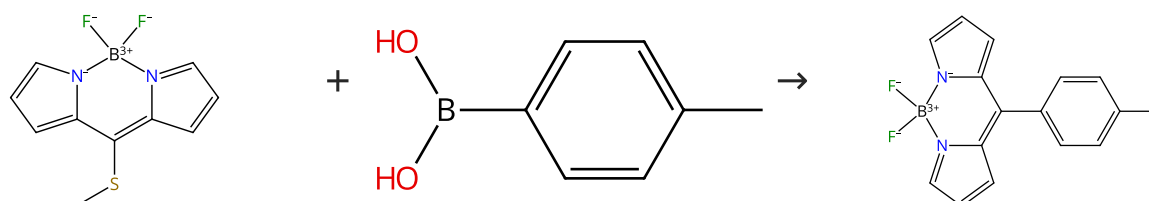
Journal of Organic Chemistry (2015), 80(11), 5771-5782.

1.1 **Reagents:** (2-Thiophenecarboxylato- $\kappa O^2, \kappa S^1$)copper
Catalysts: Tris(2-furyl)phosphine, Tris(dibenzylideneacetone) dipalladium
Solvents: Tetrahydrofuran; 5 min; 0.5 h, 55 °C

Experimental Protocols

Scheme 41 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (42)

Suppliers (114)

Suppliers (2)

31-179-CAS-18138521

Steps: 1 Yield: 93%

Scope and Limitations of the Liebeskind-Srogl Cross-Coupling Reactions Involving the Biellmann BODIPY

By: Betancourt-Mendiola, Lourdes; et al

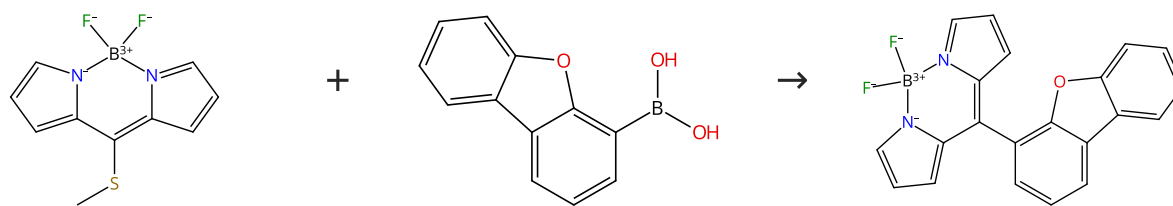
Journal of Organic Chemistry (2015), 80(11), 5771-5782.

1.1 **Reagents:** (2-Thiophenecarboxylato- $\kappa O^2, \kappa S^1$)copper
Catalysts: Tris(2-furyl)phosphine, Tris(dibenzylideneacetone) dipalladium
Solvents: Tetrahydrofuran; 5 min; 1.5 h, 55 °C

Experimental Protocols

Scheme 42 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (42)

Suppliers (92)

31-179-CAS-18194941

Steps: 1 Yield: 93%

Simple, general, and efficient synthesis of meso-substituted borondipyrrromethenes from a single platform

By: Pena-Cabrera, Eduardo; et al

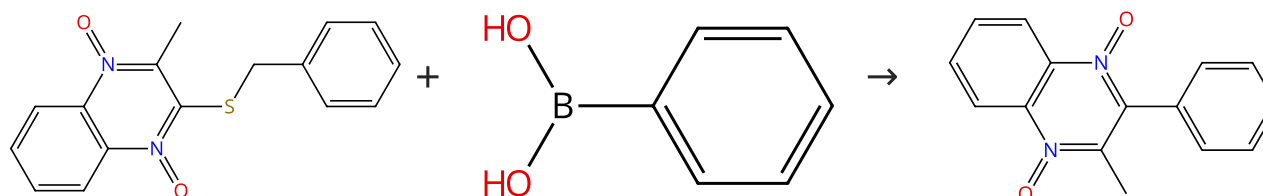
Organic Letters (2007), 9(20), 3985-3988.

1.1 **Catalysts:** Tris(2-furyl)phosphine, Tris(dibenzylideneacetone) dipalladium, (2-Thiophenecarboxylato- $\kappa O^2, \kappa S^1$)copper
Solvents: Tetrahydrofuran; 10 min, rt; 2 h, 55 °C

Experimental Protocols

Scheme 43 (1 Reaction)

Steps: 1 Yield: 93%



Supplier (1)

Suppliers (127)

Suppliers (3)

31-179-CAS-21329003

Steps: 1 Yield: 93%

Near Room Temperature Cross-Coupling Reactions of Arene Boronic Acids with a Quinoxaline 1,4-Dioxide Benzylsulfanyl Derivative

By: Dahbi, Samir; et al

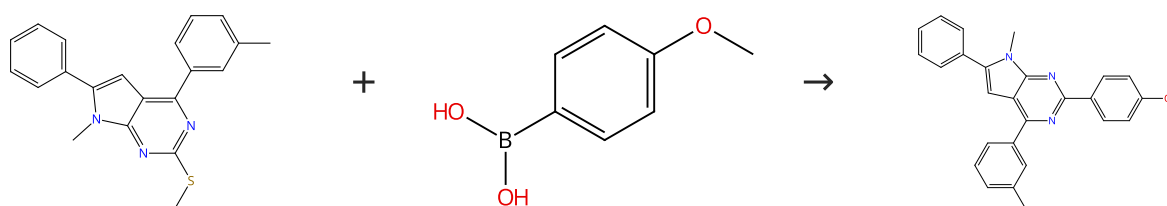
European Journal of Organic Chemistry (2012), 2012(20), 3759-3763, S3759/1-S3759/69.

1.1 **Catalysts:** Tetrakis(triphenylphosphine)palladium, (2-Thiophenecarboxylato- $\kappa O^2, \kappa S^1$)copper
Solvents: Tetrahydrofuran; 5 h, 30 °C

Experimental Protocols

Scheme 44 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (113)

31-179-CAS-21200409

Steps: 1 Yield: 93%

A Strategy for the Triarylation of Pyrrolopyrimidines by Using Microwave-Promoted Cross-Coupling Reactions

By: Prieur, Vanessa; et al

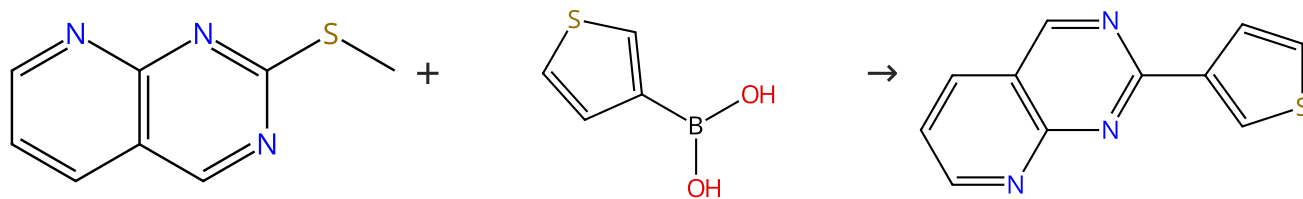
European Journal of Organic Chemistry (2015), 2015(29), 6547-6556.

1.1 **Catalysts:** Tetrakis(triphenylphosphine)palladium, [2-(Hydroxy- κO)-3-methylbenzoato- κO]copper
Solvents: Tetrahydrofuran; 1 h, 100 °C

Experimental Protocols

Scheme 45 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (3)

Suppliers (100)

Suppliers (5)

31-179-CAS-22113462

Steps: 1 Yield: 93%

Photochemical route for the synthesis of novel 2- monosubs tituted pyrido[2,3-d]pyrimidines by palladium-catalyzed cross-coupling reactions

By: Riadi, Yassine; et al

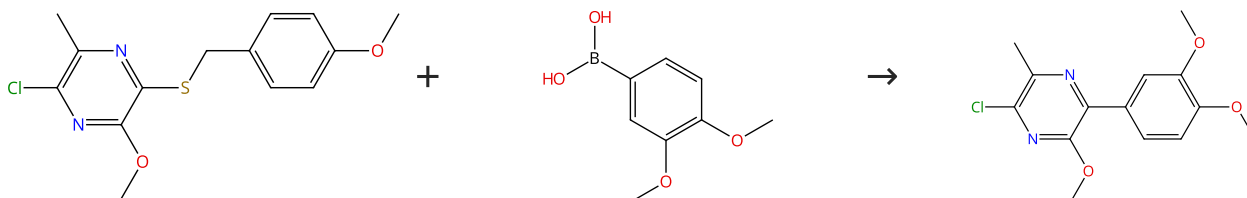
Chemical Papers (2018), 72(3), 697-701.

1.1 **Reagents:** (2-Thiophenecarboxylato- $\kappa O^2, \kappa S^1$)copper
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran; 3 h

Experimental Protocols

Scheme 46 (1 Reaction)

Steps: 1 Yield: 93%



Supplier (1)

Suppliers (104)

Supplier (1)

31-614-CAS-6972826

Steps: 1 Yield: 93%

Efficient preparation of tetrasubstituted pyrazines starting from pyrazin-2(1H)-ones

By: Modha, Sachin G.; et al

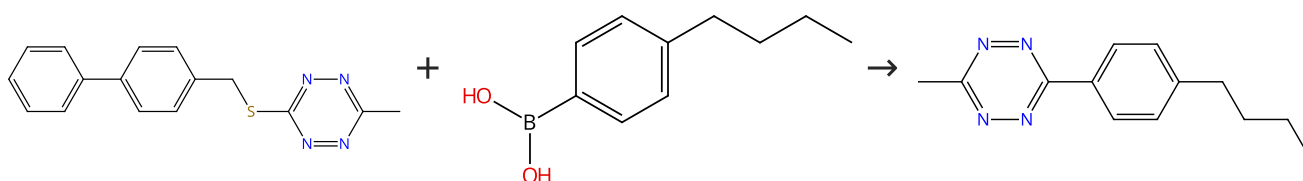
Synthesis (2012), 44(11), 1614-1624.

1.1 **Reagents:** (2-Thiophenecarboxylato- $\kappa O^2, \kappa S^1$)copper
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran; 30 min, 120 °C; 30 min, 120 °C

Experimental Protocols

Scheme 47 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (29)

Suppliers (85)

31-179-CAS-22123922

Steps: 1 Yield: 93%

Installation of Minimal Tetrazines through Silver-Mediated Liebeskind-Srogl Coupling with Arylboronic Acids

By: Lambert, William D.; et al

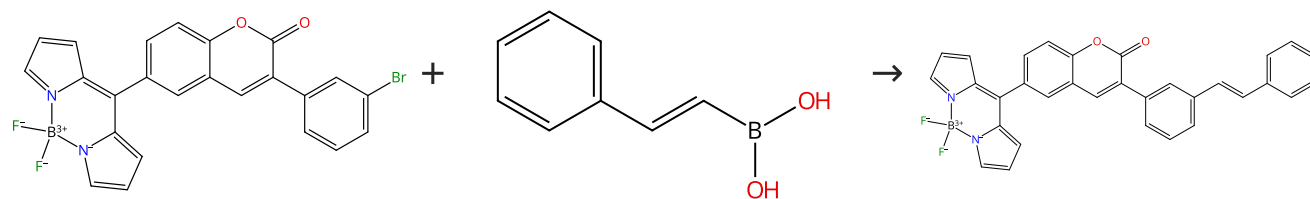
Journal of the American Chemical Society (2019), 141(43), 17068-17074.

1.1 **Reagents:** Silver oxide (Ag₂O)
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Dimethylformamide; 19 - 21 h, 60 °C

Experimental Protocols

Scheme 48 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (37)

31-177-CAS-22682954

Steps: 1 Yield: 93%

1.1 **Reagents:** Sodium carbonate, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl

Catalysts: Palladium diacetate

Solvents: Toluene, Water; 16 h, 90 °C

Experimental Protocols

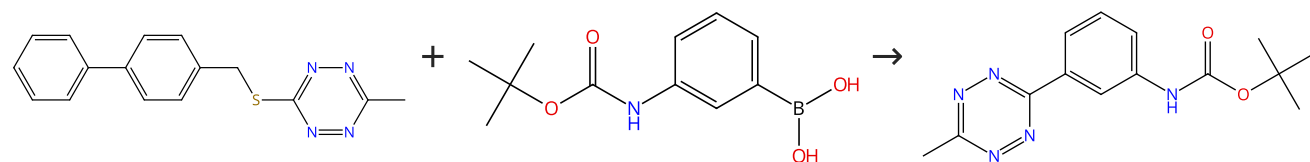
Ready access to molecular rotors based on boron dipyrromethene dyes-coumarin dyads featuring broadband absorption

By: Enriquez-Palacios, Ernesto; et al

Molecules (2020), 25(4), 781.

Scheme 49 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (29)

Suppliers (88)

31-179-CAS-22122053

Steps: 1 Yield: 93%

1.1 **Reagents:** Silver oxide (Ag₂O)

Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium

Solvents: Dimethylformamide; 19 - 21 h, 60 °C

Experimental Protocols

Installation of Minimal Tetrazines through Silver-Mediated Liebeskind-Srogl Coupling with Arylboronic Acids

By: Lambert, William D.; et al

Journal of the American Chemical Society (2019), 141(43), 17068-17074.